

Cookbook Modular

Bloques de código independientes y reutilizables

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CSS**JS**

1. Tema oscuro / claro

Sistema de temas con variables CSS, cambio por data-theme en body y persistencia en localStorage.

CSS

```
:root {
  --bg: #0d1117;
  --card: #161b22;
  --accent: #58a6ff;
  --text: #c9d1d9;
}
body[data-theme="claro"] {
```

```

--bg: #f6f8fa;
--card: #ffffff;
--accent: #0969da;
--text: #1f2328;
}
body {
  background: var(--bg);
  color: var(--text);
  transition: background 0.3s, color 0.3s;
}

```

JS

```

function initTheme() {
  const saved = localStorage.getItem('theme') || 'oscuro';
  document.body.setAttribute('data-theme', saved);
}

function toggleTheme() {
  const current = document.body.getAttribute('data-theme');
  const next = current === 'oscuro' ? 'claro' : 'oscuro';
  document.body.setAttribute('data-theme', next);
  localStorage.setItem('theme', next);
}

initTheme();

```

JS 2. Idiomas (i18n) — 25 idiomas

Sistema de traducciones con 25 idiomas, objeto de claves L, función t(), setLanguage y detección automática del idioma del navegador.

JS

```

const L = {
  es: {
    title: '🎰 Ruleta de Imágenes',
    drop: 'Arrastra imágenes aquí',
    empty: 'Agrega imágenes para comenzar',
    winner: '🎉 ¡GANADOR! 🎉',
    sound: 'Sonido'
  },
  en: {
    title: '🎰 Image Roulette',

```

```
drop: 'Drag images here',
empty: 'Add images to start',
winner: '🎉 WINNER! 🎉',
sound: 'Sound'
},
pt: {
  title: '🎡 Roleta de Imagens',
  drop: 'Arraste imagens aqui',
  empty: 'Adicione imagens para começar',
  winner: '🎉 VENCEDOR! 🎉',
  sound: 'Som'
},
fr: {
  title: '🎡 Roulette à Images',
  drop: 'Glissez des images ici',
  empty: 'Ajoutez des images pour commencer',
  winner: '🎉 GAGNANT! 🎉',
  sound: 'Son'
},
de: {
  title: '🎡 Bild-Roulette',
  drop: 'Bilder hierher ziehen',
  empty: 'Füge Bilder hinzu, um zu starten',
  winner: '🎉 GEWINNER! 🎉',
  sound: 'Ton'
},
zh: {
  title: '🎡 图片轮盘',
  drop: '将图片拖放到这里',
  empty: '添加图片开始',
  winner: '🎉 获胜者! 🎉',
  sound: '声音'
},
hi: {
  title: '🎡 इमेज रूलेट',
  drop: 'यहाँ चित्र खींचें',
  empty: 'शुरू करने के लिए चित्र जोड़ें',
  winner: '🎉 विजेता! 🎉',
  sound: 'ध्वनि'
},
ar: {
  title: '🎡 روليت الصور',
  drop: 'اسحب الصور هنا',
  empty: 'أضف الصور للبدء',
  winner: '🎉 الفائز! 🎉',
  sound: 'صوت'
},
bn: {
  title: '🎡 ইমেজ রুলেট',
```

```

drop: 'এখানে ছবি টেনে আনুন',
empty: 'শুরু করতে ছবি যোগ করুন',
winner: '🏆 বিজয়ী! 🏆',
sound: 'শব্দ'
},
ru: {
  title: '🎰 Рулетка изображений',
  drop: 'Перетащите изображения сюда',
  empty: 'Добавьте изображения для начала',
  winner: '🏆 ПОБЕДИТЕЛЬ! 🏆',
  sound: 'Звук'
},
ur: {
  title: '🎰 امیج رولیت',
  drop: 'تصاویر یہاں گھسیٹیں',
  empty: 'شروع کرنے کے لیے تصاویر شامل کریں',
  winner: '🏆 فاتح! 🏆',
  sound: 'آواز'
},
id: {
  title: '🎰 Roulette Gambar',
  drop: 'Seret gambar ke sini',
  empty: 'Tambahkan gambar untuk memulai',
  winner: '🏆 PEMENANG! 🏆',
  sound: 'Suara'
},
ja: {
  title: '🎰 画像ルーレット',
  drop: 'ここに画像をドラッグ',
  empty: '画像を追加して開始',
  winner: '🏆 勝者! 🏆',
  sound: 'サウンド'
},
pa: {
  title: '🎰 ਚਿੱਤਰ ਰੁਲੇਟ',
  drop: 'ਚਿੱਤਰ ਇੱਥੇ ਖਿੱਚੋ',
  empty: 'ਸ਼ੁਰੂ ਕਰਨ ਲਈ ਚਿੱਤਰ ਜੋੜੋ',
  winner: '🏆 ਜੇਤੂ! 🏆',
  sound: 'ਆਵਾਜ਼'
},
jv: {
  title: '🎰 Roulette Gambar',
  drop: 'Sèrèt gambar ing kéné',
  empty: 'Tambah gambar kanggo miwiti',
  winner: '🏆 PEMENANG! 🏆',
  sound: 'Swara'
},
vi: {
  title: '🎰 Vòng quay Hình ảnh',

```

```

    drop: 'Kéo hình ảnh vào đây',
    empty: 'Thêm hình ảnh để bắt đầu',
    winner: ' 🏆 NGƯỜI CHIẾN THẮNG! 🏆 ',
    sound: 'Âm thanh'
  },
  te: {
    title: ' 🎮 ಇಮೇಜ್ ರೂಲೆಟ್',
    drop: 'చిత్రాలను ఇక్కడ లాగండి',
    empty: 'ప్రారంభించడానికి చిత్రాలను జోడించండి',
    winner: ' 🏆 విజేత! 🏆 ',
    sound: 'ధ్వని'
  },
  ko: {
    title: ' 🎮 이미지 룰렛',
    drop: '이미지를 여기로 끌어다 놓으세요',
    empty: '시작하려면 이미지를 추가하세요',
    winner: ' 🏆 승자! 🏆 ',
    sound: '소리'
  },
  tr: {
    title: ' 🎮 Görsel Ruleti',
    drop: 'Resimleri buraya sürükleyin',
    empty: 'Başlamak için resim ekleyin',
    winner: ' 🏆 KAZANAN! 🏆 ',
    sound: 'Ses'
  },
  fa: {
    title: ' 🎮 رولت تصویر',
    drop: 'تصاویر را اینجا بکشید',
    empty: 'برای شروع تصاویر را اضافه کنید',
    winner: ' 🏆 برنده! 🏆 ',
    sound: 'صدا'
  },
  mr: {
    title: ' 🎮 इमेज रूलेट',
    drop: 'चित्र येथे ड्रॅग करा',
    empty: 'सुरू करण्यासाठी चित्रे जोडा',
    winner: ' 🏆 विजेता! 🏆 ',
    sound: 'आवाज'
  },
  ta: {
    title: ' 🎮 பட ரவுலட்',
    drop: 'படங்களை இங்கே இழுக்கவும்',
    empty: 'தொடங்க படங்களைச் சேர்க்கவும்',
    winner: ' 🏆 வெற்றியாளர்! 🏆 ',
    sound: 'ஒலி'
  },
  ha: {
    title: ' 🎮 Roulette na Hotuna',

```

```

    drop: 'Jawo hotuna nan',
    empty: 'Kara hotuna don farawa',
    winner: '🎉 NASARA! 🎉',
    sound: 'Sauti'
  },
  sw: {
    title: '🎲 Rulleti ya Picha',
    drop: 'Buruta picha hapa',
    empty: 'Ongeza picha kuanza',
    winner: '🎉 MSHINDI! 🎉',
    sound: 'Sauti'
  },
  it: {
    title: '🎡 Ruota delle Immagini',
    drop: 'Trascina le immagini qui',
    empty: 'Aggiungi immagini per iniziare',
    winner: '🎉 VINCITORE! 🎉',
    sound: 'Audio'
  }
};

let currentLang = localStorage.getItem('lang') || detectLang();

function detectLang() {
  const nav = navigator.language.slice(0, 2);
  return L[nav] ? nav : 'es';
}

function t(key) {
  return L[currentLang][key];
}

function setLanguage(lang) {
  if (!L[lang]) lang = 'es';
  currentLang = lang;
  localStorage.setItem('lang', lang);
  document.querySelectorAll('[data-i18n]').forEach(el => {
    const key = el.getAttribute('data-i18n');
    el.textContent = t(key) || key;
  });
}

setLanguage(currentLang);

```

JS 3. Botón de sonido

Toggle de sonido on/off con persistencia en localStorage y actualización visual del ícono.

HTML

```
<button id="soundBtn" onclick="toggleSound()">
  <span id="soundIcon">🔊</span>
</button>
```

JS

```
let soundEnabled = localStorage.getItem('sound') !== 'off';

function toggleSound() {
  soundEnabled = !soundEnabled;
  localStorage.setItem('sound', soundEnabled ? 'on' : 'off');
  updateSoundIcon();
}

function updateSoundIcon() {
  const icon = document.getElementById('soundIcon');
  if (icon) icon.textContent = soundEnabled ? '🔊' : '🔇';
}

updateSoundIcon();
```

JS 4. Web Audio - efectos

Efectos de sonido sintetizados con Web Audio API. Sin archivos externos, todo generado por código.

JS

```
let audioCtx, masterGain;

function getAudio() {
  if (!audioCtx) {
    audioCtx = new (window.AudioContext || window.webkitAudioContext)();
    masterGain = audioCtx.createGain();
    masterGain.connect(audioCtx.destination);
    masterGain.gain.value = 0.5;
  }
}
```

```
    return audioCtx;
  }

function playTick() {
  if (!soundEnabled) return;
  const ctx = getAudio();
  const osc = ctx.createOscillator();
  const gain = ctx.createGain();
  osc.connect(gain);
  gain.connect(masterGain);
  osc.frequency.value = 1800;
  osc.type = 'sine';
  gain.gain.setValueAtTime(0.3, ctx.currentTime);
  gain.gain.exponentialRampToValueAtTime(0.001, ctx.currentTime + 0.05);
  osc.start(ctx.currentTime);
  osc.stop(ctx.currentTime + 0.05);
}

function playWinSound() {
  if (!soundEnabled) return;
  const ctx = getAudio();
  [523, 659, 784, 1047].forEach((freq, i) => {
    const osc = ctx.createOscillator();
    const gain = ctx.createGain();
    osc.connect(gain);
    gain.connect(masterGain);
    osc.frequency.value = freq;
    osc.type = 'sine';
    const t = ctx.currentTime + i * 0.12;
    gain.gain.setValueAtTime(0.3, t);
    gain.gain.exponentialRampToValueAtTime(0.001, t + 0.3);
    osc.start(t);
    osc.stop(t + 0.3);
  });
}

function playWhoosh() {
  if (!soundEnabled) return;
  const ctx = getAudio();
  const bufferSize = ctx.sampleRate * 0.3;
  const buffer = ctx.createBuffer(1, bufferSize, ctx.sampleRate);
  const data = buffer.getChannelData(0);
  for (let i = 0; i < bufferSize; i++) {
    data[i] = (Math.random() * 2 - 1) * (1 - i / bufferSize);
  }
  const source = ctx.createBufferSource();
  source.buffer = buffer;
  const filter = ctx.createBiquadFilter();
  filter.type = 'bandpass';
```

```
filter.frequency.value = 800;
source.connect(filter);
filter.connect(masterGain);
source.start();
}

function playErrorSound() {
  if (!soundEnabled) return;
  const ctx = getAudio();
  const osc = ctx.createOscillator();
  const gain = ctx.createGain();
  osc.connect(gain);
  gain.connect(masterGain);
  osc.frequency.value = 200;
  osc.type = 'square';
  gain.gain.setValueAtTime(0.2, ctx.currentTime);
  gain.gain.exponentialRampToValueAtTime(0.001, ctx.currentTime + 0.4);
  osc.start(ctx.currentTime);
  osc.stop(ctx.currentTime + 0.4);
}

function playNotif() {
  if (!soundEnabled) return;
  const ctx = getAudio();
  const osc = ctx.createOscillator();
  const gain = ctx.createGain();
  osc.connect(gain);
  gain.connect(masterGain);
  osc.frequency.value = 880;
  osc.type = 'sine';
  gain.gain.setValueAtTime(0.25, ctx.currentTime);
  gain.gain.exponentialRampToValueAtTime(0.001, ctx.currentTime + 0.15);
  osc.start(ctx.currentTime);
  osc.stop(ctx.currentTime + 0.15);
}
```

HTML JS 5. Drag & Drop

Zona de arrastrar y soltar archivos con input alternativo. Manejo de eventos dragover, dragleave y drop.

CSS

```
.drop-zone {
  border: 2px dashed #58a6ff;
```

```
border-radius: 12px;
padding: 40px;
text-align: center;
cursor: pointer;
transition: background 0.2s;
}
.drop-zone.dragover {
  background: rgba(88, 166, 255, 0.1);
  border-color: #f0c040;
}
```

HTML

```
<div class="drop-zone" id="dropZone">
  <p>Arrastra archivos aquí o</p>
  <input type="file" id="fileInput" multiple
    onchange="handleFiles(this.files)">
</div>
```

JS

```
const dropZone = document.getElementById('dropZone');

dropZone.addEventListener('dragover', e => {
  e.preventDefault();
  dropZone.classList.add('dragover');
});

dropZone.addEventListener('dragleave', () => {
  dropZone.classList.remove('dragover');
});

dropZone.addEventListener('drop', e => {
  e.preventDefault();
  dropZone.classList.remove('dragover');
  handleFiles(e.dataTransfer.files);
});

function handleFiles(files) {
  for (const file of files) {
    console.log('Archivo:', file.name, file.size);
  }
}
```

JS 6. Procesamiento de imágenes

Redimensionamiento de imágenes hasta 800px y generación de miniaturas con Canvas.

JS

```
function processImage(file, maxPx = 800) {
  return new Promise(resolve => {
    const reader = new FileReader();
    reader.onload = e => {
      const img = new Image();
      img.onload = () => {
        let w = img.width, h = img.height;
        if (w > maxPx || h > maxPx) {
          if (w > h) { h = h * maxPx / w; w = maxPx; }
          else { w = w * maxPx / h; h = maxPx; }
        }
        const canvas = document.createElement('canvas');
        canvas.width = w;
        canvas.height = h;
        const ctx = canvas.getContext('2d');
        ctx.drawImage(img, 0, 0, w, h);
        resolve(canvas.toDataURL('image/jpeg', 0.85));
      };
      img.src = e.target.result;
    };
    reader.readAsDataURL(file);
  });
}

function makeThumb(file, size = 100) {
  return processImage(file, size);
}
```

JS 7. localStorage

Funciones wrapper para guardar, cargar y eliminar datos en localStorage con serialización JSON.

JS

```
function save(key, value) {
  try {
    localStorage.setItem(key, JSON.stringify(value));
  }
```

```
        return true;
    } catch (e) {
        console.error('Error guardando:', e);
        return false;
    }
}

function load(key, fallback = null) {
    try {
        const data = localStorage.getItem(key);
        return data !== null ? JSON.parse(data) : fallback;
    } catch (e) {
        console.error('Error cargando:', e);
        return fallback;
    }
}

function remove(key) {
    localStorage.removeItem(key);
}
```

JS 8. Crypto random

Generación de números aleatorios criptográficamente seguros: float, entero en rango, elección de array y shuffle Fisher-Yates.

JS

```
function cryptoRand() {
    const array = new Uint32Array(1);
    crypto.getRandomValues(array);
    return array[0] / 4294967296;
}

function cryptoInt(min, max) {
    return Math.floor(cryptoRand() * (max - min + 1)) + min;
}

function cryptoChoice(arr) {
    return arr[cryptoInt(0, arr.length - 1)];
}

function cryptoShuffle(arr) {
    const a = [...arr];
    for (let i = a.length - 1; i > 0; i--) {
```

```
    const j = cryptoInt(0, i);
    [a[i], a[j]] = [a[j], a[i]];
  }
  return a;
}
```

FIREBASE 9. Google Login

Autenticación con Google usando Firebase Auth. Login, logout, listener de estado y manejo de UI.

HTML

```
<div id="loginSection">
  <button onclick="doLogin()">Iniciar con Google</button>
</div>
<div id="userSection" style="display:none">
  <span id="userName"></span>
  <img id="userPhoto" width="40" style="border-radius:50%">
  <button onclick="doLogout()">Cerrar</button>
</div>
```

JS

```
const auth = firebase.auth();
const provider = new firebase.auth.GoogleAuthProvider();

function doLogin() {
  auth.signInWithPopup(provider).catch(err => {
    console.error('Error login:', err);
  });
}

function doLogout() {
  auth.signOut();
}

auth.onAuthStateChanged(user => {
  if (user) {
    document.getElementById('loginSection').style.display = 'none';
    document.getElementById('userSection').style.display = 'block';
    document.getElementById('userName').textContent = user.displayName;
    document.getElementById('userPhoto').src = user.photoURL;
  } else {
    document.getElementById('loginSection').style.display = 'block';
  }
});
```

```
document.getElementById('userSection').style.display = 'none';  
}  
});
```

FIREBASE 10. Firestore

Operaciones CRUD con Firestore: guardar, cargar, listar documentos del usuario y eliminar.

JS

```
const db = firebase.firestore();  
  
async function saveDoc(collection, id, data) {  
  await db.collection(collection).doc(id).set(data, { merge: true });  
}  
  
async function loadDoc(collection, id) {  
  const snap = await db.collection(collection).doc(id).get();  
  return snap.exists ? snap.data() : null;  
}  
  
async function loadUserDocs(collection, uid) {  
  const snap = await db.collection(collection)  
    .where('uid', '==', uid)  
    .orderBy('createdAt', 'desc')  
    .get();  
  return snap.docs.map(doc => ({ id: doc.id, ...doc.data() }));  
}  
  
async function deleteDoc(collection, id) {  
  await db.collection(collection).doc(id).delete();  
}
```

MEDIARECORDER 11. Grabar canvas a MP4

Grabación de canvas HTML a archivo usando MediaRecorder con captureStream. Inicio, parada y descarga.

JS

```
let mediaRecorder;
let recordedChunks = [];

function startRecording(canvas, fps = 30) {
  const stream = canvas.captureStream(fps);
  recordedChunks = [];
  mediaRecorder = new MediaRecorder(stream, {
    mimeType: 'video/webm;codecs=vp9'
  });
  mediaRecorder.ondataavailable = e => {
    if (e.data.size > 0) recordedChunks.push(e.data);
  };
  mediaRecorder.start();
}

function stopRecording() {
  return new Promise(resolve => {
    mediaRecorder.onstop = () => {
      const blob = new Blob(recordedChunks, { type: 'video/webm' });
      resolve(blob);
    };
    mediaRecorder.stop();
  });
}

async function downloadBlob(blob, filename = 'grabacion.webm') {
  const url = URL.createObjectURL(blob);
  const a = document.createElement('a');
  a.href = url;
  a.download = filename;
  a.click();
  URL.revokeObjectURL(url);
}
```

MEDIARECORDER 12. Grabar con audio

Grabación combinada de canvas + audio usando MediaStreamDestination conectado a masterGain.

JS

```
let combinedRecorder;
let combinedChunks = [];
```

```
function startRecordingWithAudio(canvas, masterGainNode, fps = 30) {
  const ctx = getAudio();
  const dest = ctx.createMediaStreamDestination();
  masterGainNode.connect(dest);

  const videoStream = canvas.captureStream(fps);
  const tracks = [
    ...videoStream.getVideoTracks(),
    ...dest.stream.getAudioTracks()
  ];
  const combinedStream = new MediaStream(tracks);

  combinedChunks = [];
  combinedRecorder = new MediaRecorder(combinedStream, {
    mimeType: 'video/webm;codecs=vp9,opus'
  });
  combinedRecorder.ondataavailable = e => {
    if (e.data.size > 0) combinedChunks.push(e.data);
  };
  combinedRecorder.start();
}

function stopRecordingWithAudio() {
  return new Promise(resolve => {
    combinedRecorder.onstop = () => {
      const blob = new Blob(combinedChunks, { type: 'video/webm' });
      resolve(blob);
    };
    combinedRecorder.stop();
  });
}
```

JS 13. Canvas dibujar/animar

Configuración de canvas 2D, dibujo de imagen circular con clip, función de easing y loop de animación con requestAnimationFrame.

JS

```
function setupCanvas(canvas, w, h) {
  canvas.width = w;
  canvas.height = h;
  return canvas.getContext('2d');
}
```

```
function drawImageCircle(ctx, img, cx, cy, radius) {
  ctx.save();
  ctx.beginPath();
  ctx.arc(cx, cy, radius, 0, Math.PI * 2);
  ctx.clip();
  const size = radius * 2;
  ctx.drawImage(img, cx - radius, cy - radius, size, size);
  ctx.restore();
}

function easeOut(t) {
  return 1 - Math.pow(1 - t, 3);
}

function animateLoop(callback) {
  let startTime = null;
  function loop(timestamp) {
    if (!startTime) startTime = timestamp;
    const elapsed = timestamp - startTime;
    callback(elapsed, timestamp);
    requestAnimationFrame(loop);
  }
  requestAnimationFrame(loop);
}
```

JS 14. Firma by SandovalJon

Pie de página con firma del autor.

HTML

```
<footer style="text-align:center;color:#484f58;
  margin-top:40px;padding:20px;font-size:0.85rem;">
  by SandovalJon
</footer>
```